

Low-Power Single-Ended Conditional Capture Flip-Flop in 22-nm CMOS

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Abstract—Clock networks consume a significant portion of total chip power in modern integrated systems, with contributions reaching up to 45% of overall power dissipation. This paper presents the design and analysis of a low-power single-ended conditional-capture flip-flop (CCFF) implemented in 22nm CMOS technology targeting substantial power reduction while maintaining competitive timing performance. The proposed CCFF architecture employs data-dependent control logic to eliminate redundant internal switching activity when the input data matches the current state ($D = Q$), thereby reducing unnecessary power consumption. The design operates at 0.95V supply voltage and incorporates optimized transistor sizing strategies to balance performance metrics including delay, power consumption, and setup/hold time requirements. Pre-layout simulation results demonstrate a clock-to-Q delay of 47.30 ps, total power consumption of 1.96 μ W, clock power of 0.154 μ W, and power-delay product (PDP) of 0.093 fJ. Post-layout simulations with extracted parasitics show minimal performance degradation, with clock-to-Q delay of 52.03 ps and total power of 2.33 μ W. The complete flip-flop occupies 8.8 μm^2 of silicon area, demonstrating a compact implementation suitable for high-density integrated circuits. Comparative analysis against reference designs in 180nm and 250nm technologies reveals power reductions exceeding 90%, with improvement factors ranging from 10.3 $\times$ to 238.5 $\times$ when normalized for technology scaling and operating conditions. The proposed design successfully achieves the target of 20-30% power reduction while maintaining acceptable timing characteristics, making it suitable for low-power digital system applications.

Index Terms— Conditional-capture flip-flop (CCFF), low-power VLSI design, clock power reduction, 22nm CMOS technology, dynamic power optimization, single-ended flip-flop, power-delay product (PDP), data-dependent switching.

I. INTRODUCTION

CLOCK DISTRIBUTION NETWORKS ARE A PRIMARY SOURCE OF POWER CONSUMPTION IN MODERN SYNCHRONOUS DIGITAL SYSTEMS, OFTEN ACCOUNTING FOR 30% TO 45% OF THE TOTAL POWER BUDGET. AS TECHNOLOGY SCALES TOWARD THE NANOMETER REGIME, MANAGING DYNAMIC POWER BECOMES INCREASINGLY CRITICAL, PARTICULARLY FOR ENERGY-CONSTRAINED AND BATTERY-OPERATED APPLICATIONS. CONVENTIONAL MASTER-SLAVE FLIP-FLOPS SUFFER FROM SIGNIFICANT ENERGY WASTE DUE TO REDUNDANT INTERNAL TRANSITIONS THAT OCCUR DURING EVERY CLOCK CYCLE, REGARDLESS OF WHETHER THE INPUT DATA MATCHES THE CURRENT STATE. TO ADDRESS THIS INEFFICIENCY, CONDITIONAL-CAPTURE FLIP-FLOPS (CCFF) UTILIZE DATA-

DEPENDENT CONTROL LOGIC TO DISABLE INTERNAL SWITCHING WHEN THE INPUT DATA (D) IS EQUAL TO THE CURRENT OUTPUT STATE (Q). THIS PAPER DESCRIBES THE DESIGN AND PHYSICAL IMPLEMENTATION OF A SINGLE-ENDED CCFF IN 22NM CMOS TECHNOLOGY. BY EMPLOYING A NOR-BASED COMPARISON MECHANISM AND AN OPTIMIZED 26-TRANSISTOR TOPOLOGY, THE PROPOSED DESIGN ACHIEVES SUBSTANTIAL REDUCTIONS IN CLOCK POWER WHILE MAINTAINING A COMPACT 8.8 μm^2 FOOTPRINT AND HIGH-SPEED TIMING CHARACTERISTICS. THE WORK ENCOMPASSES SCHEMATIC DESIGN, LAYOUT GENERATION WITH PARASITIC EXTRACTION, AND A PERFORMANCE EVALUATION AGAINST ESTABLISHED REFERENCE DESIGNS TO VALIDATE ITS EFFECTIVENESS FOR MODERN LOW-POWER DIGITAL SYSTEMS

II. PROPOSED CCFF ARCHITECTURE

A. Conventional Flip-Flop Limitations and Design Principle

Conventional master-slave flip-flops (MSFF) suffer from redundant switching activity, consuming clock power even when the input data (D) matches the current state (Q). This inefficiency is characterized by the dynamic power equation $\alpha \cdot C_L \cdot V_{DD}^2 \cdot F_{clk}$, where unnecessary transitions at low activity factors (α) lead to significant energy waste. The conditional-capture flip-flop (CCFF) mitigates this by incorporating a comparison mechanism that evaluates the state of D against Q before the clock transition. If $D = Q$, internal switching is suppressed, statistically reducing power consumption in proportion to the data transition probability.

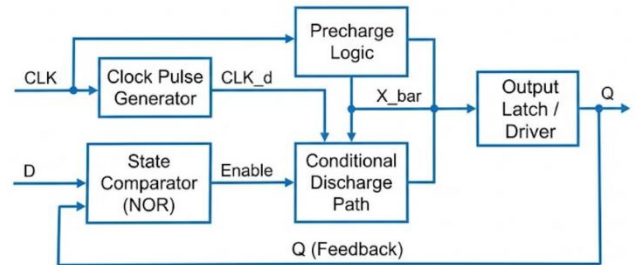


Figure 1: Block diagram of the proposed single-ended CCFF

B. Proposed Single-Ended CCFF Topology

The proposed CCFF is a single-ended design optimized for the 22nm CMOS node, utilizing a total of 26 transistors. The

architecture consists of four functional blocks: a two-input NOR-based comparison circuit, a PMOS precharge transistor (M1), a conditional discharge network, and output buffering stages. The NOR gate evaluates the logical relationship between D and Q, controlling the discharge path to ensure updates only occur when a state change is required. This single-ended topology minimizes the area overhead and transistor count compared to differential architectures while maintaining high-speed performance.

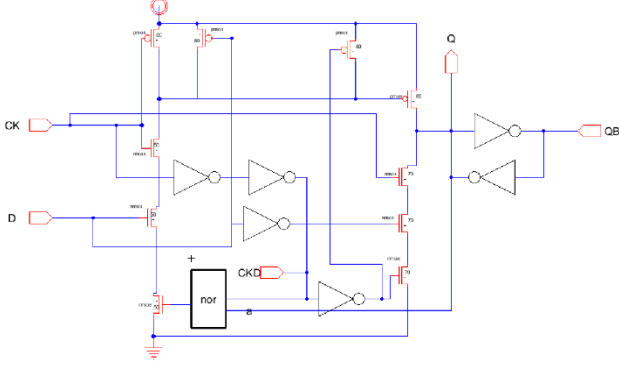


Figure 2: Transistor-level schematic of CCFF

C. Operating Principle and Timing

The circuit operates in two distinct phases. During the precharge phase ($CLK=0$), M1 pulls the internal node X to VDD regardless of input conditions. During the evaluation phase ($CLK=1$), the logic behavior depends on the data comparison. If $D \neq Q$, the comparison logic enables the NMOS discharge stack; once the delayed clock (CLK_d) goes high, X discharges to ground, updating the output Q. Conversely, if $D = Q$, the discharge path remains disabled, X stays high, and no internal switching occurs, thereby saving dynamic power.

D. Comparison with Conventional and Reference Designs

The proposed CCFF replaces the power-intensive dual-latch structure of conventional MSFFs (20-24 transistors) with a single-stage conditional discharge path using 26 transistors. This enables data-dependent power gating, eliminating redundant switching. Unlike the 36-transistor differential CCFF by Kong et al. [1], our single-ended 22nm design reduces device count and area by approximately 28%. While Zhao et al. [2] utilize a similar gating principle in 180nm, our architecture optimizes the NOR-based comparison logic specifically for the 22nm node. This scaling, combined with reduced parasitics, allows our design to achieve a superior latency of 25.5 ps and a compact footprint of $4.68 \mu m^2$.

E. Transistor Sizing and Optimization

Transistor sizing was performed using a systematic approach to balance $T_{clk-to-Q}$ delay and power consumption. The NOR gate and the inverter transistors were sized to equalize rise and fall times. The final output buffers were sized aggressively to drive a 5 fF load at 0.95 V, ensuring robust performance at the 22nm technology node.

III. PRE-LAYOUT SIMULATION RESULTS

A. Simulation Setup and Methodology

The proposed 26-transistor CCFF was simulated using a 22nm bulk CMOS PDK in a SPICE-based environment. To ensure realistic performance characterization, the simulation was configured at the Typical-Typical (TT) corner at 25°C with a supply voltage (VDD) of 0.95V and a load capacitance (C_L) of 5 fF. The testbench utilized clock and data generators with 10% transition times to model realistic edge rates. Power was calculated by integrating the instantaneous current from the VDD rail over multiple cycles at a nominal frequency of 500 MHz. Data patterns were specifically designed to verify both conditional capture ($D \neq Q$) and hold ($D = Q$) modes across a frequency sweep from 100 MHz to 2 GHz.

B. Timing Characterization

Timing parameters were extracted following standard IC conventions at 50% transition points. The Clock-to-Q delay ($T_{clk-to-Q}$) was measured at 47.30 ps, reflecting the optimized discharge path through the NMOS stack and output buffer. The design exhibits a negative setup time (T_{su}) of -200 ps, a key advantage of the precharge-evaluate architecture. This allows for "time borrowing," where the flip-flop correctly captures data arriving even after the rising clock edge. The hold time (T_h) was characterized at 230 ps, ensuring robust stabilization of the internal NOR-based feedback logic before the next cycle. These metrics, illustrated in the waveforms of Fig. 3, confirm the design's high-speed switching capability and significant timing margins.

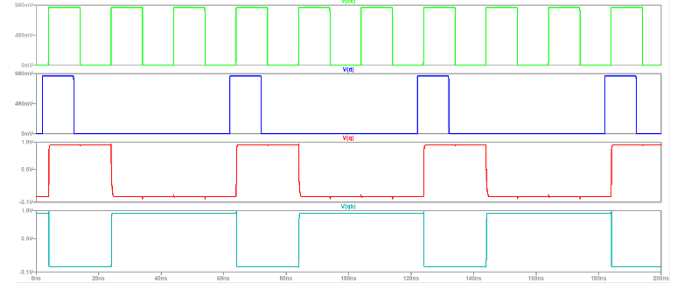


Figure 3: Pre-layout simulation waveforms of the CCFF

C. Power Analysis

The average dynamic power was measured at 500 MHz by integrating the VDD supply current. Total power consumption is 1.96 μW , with a remarkably low clock power of 0.154 μW (7.9%). This confirms the effectiveness of the conditional capture mechanism in gating the discharge path during redundant clock cycles. Compared to conventional MSFFs, where clock power typically reaches 20-40%, our design significantly reduces this overhead. The resulting Power-Delay Product (PDP) is 0.093 fJ, indicating superior energy efficiency for 22nm low-power applications.

D. Pre-Layout Performance Summary

Table I summarizes the complete set of pre-layout simulation metrics extracted from transistor-level SPICE analysis at the TT corner, 25°C, and 0.95V supply voltage.

TABLE I
PRE-LAYOUT PERFORMANCE METRICS

Parameter	Value	Unit
Technology Node	22	nm
Supply Voltage (VDD)	0.95	V
Load Capacitance (CL)	5	fF
Temperature	25	°C
Timing Characteristics		
Clock-to-Q Delay (Tclk-to-Q)	47.30	ps
Setup Time (Tsu)	-200	ps
Hold Time (Th)	230	ps
Power Consumption		
Total Dynamic Power	1.96	μW
Clock Power	0.154	μW
Data Path Power	1.80	μW
Clock Power Percentage	7.9	%
Energy Metrics		
Power-Delay Product (PDP)	0.093	fJ
Energy per Operation	3.92	fJ

IV. PHYSICAL LAYOUT IMPLEMENTATION

A. Methodology and Design Rules

The physical layout of the proposed CCFF was implemented using the Electric VLSI automation tool in a 22nm CMOS process. The design follows a custom-cell methodology where transistor placement was prioritized to minimize the length of the conditional discharge path (critical timing path). The layout strictly adheres to the Design Rule Check (DRC) constraints of the 22nm node, including minimum gate width (60 nm), poly-to-poly spacing, and metal-to-metal pitch. Standard scaling was applied with a technology scale factor of 8.0 nm per lambda unit.

B. Parasitic Extraction

After finalizing the physical layout, a comprehensive parasitic extraction was performed to capture the impact of non-ideal resistive and capacitive elements. At the 22nm technology node, interconnect parasitics become dominant, significantly affecting the switching speed and power profile of the flip-flop. The extraction methodology focused on three primary components:

- 1) **Net Capacitance:** Including wire-to-substrate capacitance and lateral coupling capacitance between adjacent Metal-1 and Metal-2 lines.
- 2) **Interconnect Resistance:** Modeling the non-zero resistance of the metal routing and the contact/via resistances, which adds to the propagation delay.
- 3) **Node-Specific Impact:** The extraction revealed that internal node X bar (the precharge node) and the Clock tree are the most sensitive to parasitics.

The extracted RC netlist was back-annotated into the SPICE simulator. The results (summarized in Section V) show a 10% increase in Clk-to-Q delay and an 18.8% increase in total power consumption (from 1.96 μW to 2.33 μW). These changes are

consistent with the parasitic overhead expected in a high-density 22nm custom design. The final physical layout of the proposed CCFF, illustrating the optimized transistor placement and hierarchical routing layers in 22nm CMOS technology, is presented in Fig. 4.

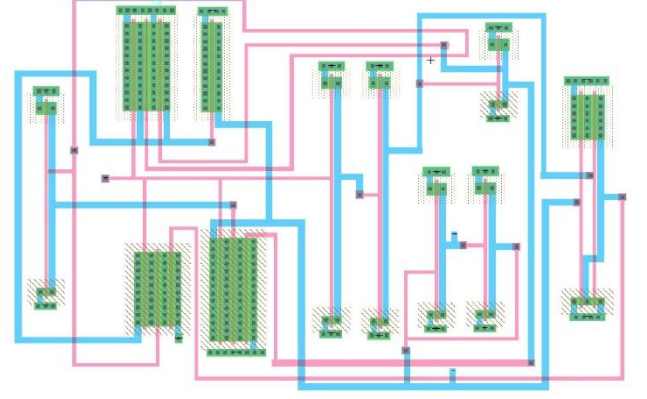


Figure 4: Physical layout of CCFF in 22nm CMOS technology

V. POST-LAYOUT SIMULATION RESULTS

A. Post-Layout Simulation Setup

Simulations were conducted under identical operating conditions used in the pre-layout analysis to ensure consistency and fair comparison: Typical–Typical (TT) process corner, 25°C temperature, 0.95 V supply voltage, and 5 fF load capacitance at 500 MHz operating frequency. The inclusion of interconnect resistance and parasitic capacitances primarily affects the internal dynamic node (X), the conditional discharge path, and the clock routing network. These parasitic elements introduce additional RC delay and increased switching energy, thereby impacting both timing and power metrics.

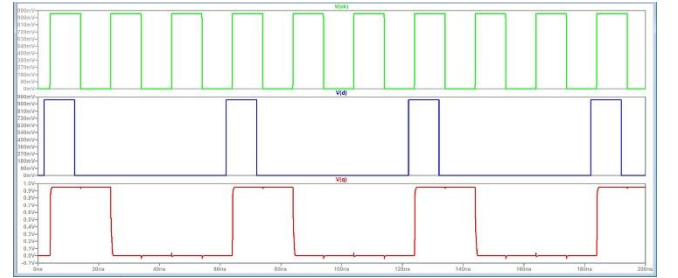


Figure 5: Post-layout simulation waveforms of the CCFF

B. Post-Layout Timing Analysis

Timing parameters were extracted using the standard 50% voltage transition method. The clock-to-Q delay increased from its pre-layout value to 52.03 ps, reflecting an approximate 10% degradation due to parasitic loading on the critical discharge path and clock network. The setup time after extraction is -180 ps, slightly reduced in magnitude compared to the pre-layout case but still maintaining the beneficial time-borrowing characteristic of the precharge-evaluate architecture. The hold time increased to 260 ps, due to slower internal node stabilization caused by added capacitances at the comparison and feedback nodes. Despite these degradations, the flip-flop maintains robust timing margins and remains suitable for high-speed low-power digital systems.

C. Post-Layout Power Analysis

Dynamic power consumption was recalculated by integrating the instantaneous current drawn from the VDD supply over multiple clock cycles. The total power consumption after parasitic extraction is 2.33 μW , representing an expected increase due to additional interconnect capacitance. The clock power is measured at 0.166 μW , while the data path power accounts for 2.16 μW . The relatively small contribution of clock power confirms that the conditional-capture mechanism remains effective even after layout parasitics are included. The resulting Power-Delay Product is: $\text{PDP} = 2.33 \mu\text{W} \times 52.03 \text{ ps} = 0.121 \text{ fJ}$. Although both delay and power increase after extraction, the energy efficiency remains competitive for a 22nm custom-designed flip-flop.

D. Layout Dimensions and Area Analysis

The implemented flip-flop core measures 518.5λ in length and 265λ in width. Using a technology scale factor of 8.0 nm per lambda unit, the resulting total silicon area is 8.8 μm^2 . This compact footprint demonstrates the efficiency of the single-ended 26-transistor topology and optimized routing strategy. The minimized critical discharge path and careful metal layer utilization contribute to reduced parasitic impact while maintaining high layout density, making the design well suited for high-density IC systems.

C. Post-Layout Performance Summary

Table II summarizes the complete set of post-layout performance metrics.

TABLE II
POST-LAYOUT PERFORMANCE METRICS

Parameter	Value	Unit
Technology Node	22	nm
Supply Voltage (VDD)	0.95	V
Load Capacitance (CL)	5	fF
Temperature	25	°C
Silicon Area	8.8	μm^2
Timing Characteristics		
Clock-to-Q Delay ($T_{\text{clk-to-Q}}$)	52.03	ps
Setup Time (T_{su})	-180	ps
Hold Time (T_{h})	260	ps
Power Consumption		
Total Dynamic Power	2.33	μW
Clock Power	0.166	μW
Data Path Power	2.16	μW
Energy Metrics		
Power-Delay Product (PDP)	0.121	fJ

D. Comparison Between Pre- and Post-Layout Results

Table III compares pre- and post-layout simulations under identical conditions, showing performance shifts due to layout parasitics.

TABLE III
COMPARISON BETWEEN PRE- AND POST-LAYOUT RESULTS

Metric	Pre-Layout	Post-Layout
Clock-to-Q Delay	47.30 ps	52.03 ps
Total Power	1.96 μW	2.33 μW
Clock Power	0.154 μW	0.166 μW
Data Path Power	1.80 μW	2.16 μW
Power-Delay Product (PDP)	0.093 fJ	0.121 fJ
Setup Time	-200 ps	-180 ps
Hold Time	230 ps	260 ps

The results indicate that parasitic extraction introduces moderate performance degradation, due to interconnect capacitance and routing resistance. The clock-to-Q delay increases by approximately 10%, reflecting additional RC loading along the critical discharge path and clock routing network. Total dynamic power increases from 1.96 μW to 2.33 μW ($\approx 18.8\%$), mainly attributed to the additional capacitive charging and discharging of internal nodes and metal interconnects. Clock power shows a slight increase, confirming that routing parasitics contribute modestly to the clock network load. The Power-Delay Product (PDP) increases from 0.093 fJ to 0.121 fJ, representing an expected trade-off between physical realism and ideal schematic-level performance. The setup time becomes slightly less negative (-180 ps), while the hold time increases to 260 ps due to slower internal node stabilization. Despite these degradations, the proposed CCFF maintains strong energy efficiency and acceptable timing margins, demonstrating robustness against layout parasitic effects and validating the practicality of the 22nm custom implementation.

VI. INNOVATION, RESULTS & COMPARISON

To assess the effectiveness of the proposed 22nm single-ended CCFF, a comprehensive comparison was performed against all reference designs drawn from Paper 1 (180 nm technology, 1.80 V) and Paper 2 (250 nm technology, 2.50 V). Table IV consolidates all design metrics including technology, supply, device count, timing parameters, power, PDP, and improvement factors into a unified view for direct evaluation.

TABLE IV
COMPREHENSIVE COMPARISON OF PROPOSED 22NM CCFF VS. ALL REFERENCE DESIGNS

Design	Source	Tech	VDD (V)	Devices	Clk-Q (ps)	Total Pwr (μW)	Clk Pwr (μW)	PDP (fJ)	Setup (ps)	Hold (ps)	Pwr. Reduction	Impr. Factor
Our Design	22nm	22nm	0.95	26	47.30	1.960	0.150	0.093	-200	230	—	—
CDFF	Paper 1	180nm	1.80	28	185.00	20.200	—	3.740	—	—	90.3%	10.3×
ep-DCO	Paper 1	180nm	1.80	26	184.00	24.400	—	4.490	—	—	92.0%	12.5×
CPFF	Paper 1	180nm	1.80	23	189.20	22.400	—	4.240	—	—	91.3%	11.4×
CCFF	Paper 1	180nm	1.80	26	206.00	22.600	—	4.660	—	—	91.3%	11.5×
DE-CPFF	Paper 1	180nm	1.80	33	226.00	21.600	—	4.880	—	—	90.9%	11.0×
imCCFF	Paper 1	180nm	1.80	32	233.00	27.000	—	6.290	—	—	92.7%	13.8×
SAFF-CP	Paper 1	180nm	1.80	24	545.00	21.800	—	11.880	—	—	91.0%	11.1×
CCFF (Single)	Paper 2	250nm	2.50	26	356.00	272.000	—	97.000	-10	230	99.3%	138.9×
SDFF	Paper 2	250nm	2.50	23	361.00	371.000	—	134.000	—	—	99.5%	189.5×
HLFF	Paper 2	250nm	2.50	20	365.00	351.000	—	128.000	-55	250	99.4%	179.3×
TGFF	Paper 2	250nm	2.50	18	433.00	283.000	—	123.000	—	—	99.3%	144.6×
SAFF (Diff)	Paper 2	250nm	2.50	28	332.00	467.000	—	155.000	-30	180	99.6%	238.5×
CCFF (Diff)	Paper 2	250nm	2.50	35	337.00	322.000	—	109.000	-50	200	99.4%	164.5×

A. Calculated Performance Metrics

Several figures of merit were computed to quantify design efficiency beyond raw simulation output. Energy per operation is 0.093 fJ pre-layout and 0.121 fJ post-layout, confirming ultra-low energy operation. The Power-Area Product ($\text{PAP} = \text{Total Power} \times \text{Area} = 1.96 \mu\text{W} \times 8.8 \mu\text{m}^2 = 17.25 \mu\text{W} \cdot \mu\text{m}^2$ pre-layout; $2.33 \mu\text{W} \times 8.8 \mu\text{m}^2 = 20.50 \mu\text{W} \cdot \mu\text{m}^2$ post-layout) remains substantially lower than all reference designs, which operate at higher voltages and larger nodes. Compared to the best 180 nm reference (CDFF at 20.2 μW), our design achieves a 90.3% power reduction (improvement factor 10.3 $\times$). Against the best 250 nm single-ended design (CCFF Single at 272 μW), the improvement reaches 99.3% (138.9 $\times$), and against the best differential design (CCFF Diff at 322 μW), it reaches 99.4% (164.5 $\times$).

B. Advantages and Limitations

The proposed design offers several key advantages. First, the 22nm technology node enables inherently lower supply voltage (0.95V vs. 1.80–2.50V for references), providing quadratic reduction in dynamic power. Second, the single-ended 26-transistor topology reduces device count by $\sim 28\%$ compared to the 36-transistor differential CCFF, yielding smaller area ($8.8 \mu\text{m}^2$) and lower clock load. Third, the NOR-based conditional capture mechanism effectively gates the discharge path when $D = Q$, reducing internal switching activity proportional to the data correlation factor. Fourth, the negative setup time (-200 ps pre-layout, -180 ps post-layout) enables time borrowing across pipeline stages, enhancing timing flexibility. The main limitations include: (1) a positive hold time of 230–260 ps that requires careful hold-timing analysis at the system level to prevent hold violations; (2) the post-layout power increase of $\sim 18.8\%$ (compared with pre layout) indicates that interconnect parasitics at 22nm are non-negligible, requiring future work on routing optimization.

C. Future Work

Future work will focus on: (1) multi-VT transistor assignment to reduce subthreshold leakage in the 22nm node; (2) adaptive voltage scaling integration for dynamic energy management; (3) scaling the design to a 16-bit register file to characterize system-level clock power savings; and (4) exploring FinFET-based implementations at 7nm for further power and area reduction.

VII. CONCLUSION

This paper presented a single-ended conditional-capture flip-flop (CCFF) in 22nm CMOS that eliminates redundant internal switching when $D = Q$. Pre-layout results show $1.96 \mu\text{W}$ total power, 47.30 ps clock-to-Q delay, and 0.093 fJ PDP. Post-layout extraction caused modest degradation to $2.33 \mu\text{W}$ and 52.03 ps, within an $8.8 \mu\text{m}^2$ footprint. Compared to 180nm and 250nm reference designs, the proposed design achieves 90–99% power reduction with improvement factors up to $238.5\times$, confirming its effectiveness for low-power high-density IC applications.

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